Sprint 4 Report (Week 1 - Week 6)

YouTube Link

GitHub Link: https://github.com/KieranMcHugh/WSUDAS-AI.git

What's New (User Facing)

The customers of WSUDAS are excited about a new predicting model for Lethal Tempo of grapes in North America: the NYUS 2.2 by a team in Cornell University. This model is capable of extracting the data from a .csv file with just the date, min and max temperature and predict the expected LT50 (lethal temperature 50%) for that date. It was shipped trained, so our task is to dive more into it and see how it is performing for integration purposes.

Work Summary (Developer Facing)

The bulk of the work for this sprint was exploring the viability of integrating the NYUS2.2 model into WSUDAS' existing service with the goal of being able to better inform grape growers in British Columbia about the likelihood of their crops' survival. This meant first learning the basics of using an autogluon model and then determining what data is needed for it to run successfully. Once we better understood these two aspects we then tested the model using historical sample data from WSUDAS to ensure the output with their data was consistent with the dataset provided in the NYUS2.2 model.

Unfinished Work

As we have started work on a separate project from what we were working on before there is a decent amount of work ahead of us. However, at this time we are where we want to be in terms of current progress on the project. We have a better understanding of the model we are working with and are now exploring ways WSUDAS can utilize it to provide enhanced services to their users. Specifically, we are looking into whether we can connect this model to WSUDAS via an API endpoint.

Completed Issues/User Stories

Here are links to the issues that we completed in this sprint:

* Depickling Python Files:
https://github.com/users/KieranMcHugh/projects/3/views/1?pane=issue&itemId=157956505&issue=KieranMcHugh%7CWSUDAS-AI%7C14
* Reformat extraction code to accept WSUDAS' data:
https://github.com/users/KieranMcHugh/projects/3/views/1?pane=issue&itemId=157956201&issue=KieranMcHugh%7CWSUDAS-AI%7C11

Code Files for Review
Please review the following code files, which were actively developed during this sprint, for quality:
*[Feature_extraction.R]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/model/Feature_extraction.R
*[WEL2_extract.ipynb]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/sample_code/WEL2_extract.ipynb
*[sample.ipynb]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/sample_code/sample.ipynb

Retrospective Summary
Here's what went well:
* Item 1: We have a good understanding of how the model works and what is needed to integrate NYUS2.2 with WSUDAS' existing system
* Item 2: We have identified drawbacks with the current form of the model and are working on ways we can address those for the production environment
* Item 3: We have successfully used both the R and Python components of the model and have tested that they behave as expected
* Item 4: We have implemented testing strategies to ensure that the model behaves as expected when passing WSUDAS' data through it, ensuring integration with WSUDAS is feasible should the client choose to move forward with this model.
Here's what we'd like to improve:
* Item 1: Our understanding of some outstanding components of the model (since the original creators of the model were not developers there is sparse documentation, so we still need to look further into certain areas to get a full understanding of how they work).
* Item 2: We need to learn more about fine tuning autogluon specifically, as while we have a grasp of the model, it offers few methods for tuning the model manually.
* Item 3: Some of the model's outputs are difficult to understand, and due to the lack of documentation and the blackbox nature of autogluon, there are still some minor aspects of the model's output that we need to clarify. While we have a decently informed idea of what some of those outputs may represent, we do not want to misinform our client about the capabilities of the model and thus need to look further into this.

Here are changes we plan to implement in the next sprint:
* Convert R code to Python code
* Start the training process on our internal data instead of using their sample data